The Arrow of Time as Petz Recovery Failure

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The Petz recovery map plays a central role across quantum information, error correction, and thermodynamics—a fact well recognized by these communities. Here we offer a physical interpretation of this mathematical unity: we define a temporal asymmetry parameter $\tau = 1 - F$, where F is the fidelity of Petz recovery, and show that τ continuously quantifies the arrow of time. For closed systems $\tau = 0$: forward and reverse processes are indistinguishable. For open systems $\tau > 0$, bounded by entropy production via $\tau \leq 1 - \exp(-\Sigma/2)$ (a consequence of the universal recovery bound of Junge et al.). This interpretation unifies the quantum eraser ($\tau = 0$ because the signal-idler pair is closed), quantum error correction ($\tau > 0$ sets the decoder's task), and thermodynamic irreversibility (τ bounded by dissipation) under a single physical picture. We present one new result: the amplitude damping channel exactly saturates the recovery bound ($F^2 = \exp(-\Delta D)$) for fixed-point states, making the arrow of time a tight information-theoretic quantity for this experimentally relevant noise model.

INTRODUCTION

The Petz recovery map [3, 4] plays a central role across several areas of quantum information science: as a near-optimal quantum error correction (QEC) decoder [5]; as the *unique* retrodiction functor in categorical quantum mechanics [6]; as the reverse channel in quantum fluctuation theorems [7]; as a physical reversal via dilations [17]; and in entanglement wedge reconstruction in holography [8]. These connections are well established in the literature, and recent work by Buscemi, Fullwood, and Parzygnat [31] has further developed the information-theoretic foundations linking them.

In this Letter, we offer a physical interpretation of this mathematical unity. We define a temporal asymmetry parameter $\tau = 1 - F$, where F is the Petz recovery fidelity, and argue that τ provides a continuous, operational measure of the arrow of time.

The interpretive framework rests on known results assembled into a single picture: (1) the quantum eraser [1, 2] is a physical instantiation of Petz recovery; (2) the uniqueness of the Petz map as retrodiction functor (due to Parzygnat–Buscemi [6]) explains why it also serves as a near-optimal QEC decoder [5, 11]; (3) the universal recovery bound of Junge et al. [11] (Corollary 5.1 therein) and the Fawzi–Renner inequality [12], combined with the Kolchinsky–Wolpert decomposition [13] and the Reeb–Wolf bound [18], yield a chain linking recovery fidelity to entropy production. The logical flow is:

$$\text{Eraser} \xrightarrow{\text{Petz}} \text{Retrodiction} \to \text{QEC} \to \text{Thermo}. \qquad (1)$$

The physical content is that the arrow of time—the empirical asymmetry between past and future—can be identified with Petz recovery failure. For a closed system, the Petz map recovers the initial state exactly ($F = 1$, $\tau = 0$); there is no distinction between forward and reverse processes. Openness to an environment ($\Sigma > 0$) breaks this symmetry, and $\tau \leq 1 - \exp(-\Sigma/2)$ quantifies the cost.

We also present one new result: we show that the amplitude damping channel exactly saturates the Junge et al. bound ($F^2 = \exp(-\Delta D)$) for fixed-point input states, while unital channels generically do not. This identifies a physically meaningful distinction between channel families in terms of temporal asymmetry.

THE PETZ RECOVERY MAP

Given a quantum channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ and a faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the *Petz recovery map* is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\, \mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\, \rho\, \mathcal{N}(\sigma)^{-1/2}\big)\, \sigma^{1/2}, \qquad (2)$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint of $\mathcal{N}$. The map $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive and trace-preserving (CPTP) [3], and satisfies exact recovery $\mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of σ if and only if the data-processing inequality (DPI) for relative entropy is saturated: $D(\sigma\|\tau) = D(\mathcal{N}(\sigma)\|\mathcal{N}(\tau))$ for all τ [4].

Table I shows how the mathematical ingredients of Eq. (2) map to physical meaning in each domain.

TABLE I. Physical interpretation of the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ across four domains.

	Eraser	**Time Rev.**	**QEC**	**Thermo.**
$\mathcal{N}$	Path mark	Fwd. evol.	Noise	Irrev. proc.
σ	Entangled	Prior	Code state	Gibbs
$\mathcal{R}$	Erasure	Retrodiction	Decoder	Rev. proc.
ΔD	Decoherence	Info. loss	Code dist.	βW_{diss}

UNIQUENESS OF RETRODICTION

We recall the uniqueness result of Parzygnat and Buscemi [6], which is central to our interpretation.

Theorem 1 (Retrodiction Uniqueness (Parzygnat–Buscemi [6])). *Let C be the category of finite-dimensional C^*-algebras with state-preserving CP maps as morphisms. A* retrodiction functor *is a contravariant functor $R : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$ satisfying:*

(i) Preservation: *R sends $(\mathcal{N}, \sigma)$ to a CPTP map.*

(ii) Bayesian consistency: *For composable channels $\mathcal{N}_1, \mathcal{N}_2$ and prior σ,*

$$R(\mathcal{N}_2 \circ \mathcal{N}_1,\, \sigma) = R(\mathcal{N}_1,\, \sigma) \circ R(\mathcal{N}_2,\, \mathcal{N}_1(\sigma)). \tag{3}$$

(iii) Normalization: $R(\mathrm{id}, \sigma) = \mathrm{id}$.

(iv) Classical limit: *On commutative subalgebras, R reduces to classical Bayes' rule.*

Then $R(\mathcal{N}, \sigma) = \mathcal{R}_{\sigma,\mathcal{N}}$ is the unique such functor.

Proof sketch.—Axioms (ii) and (iv) force the $\sigma^{1/2}(\cdot)\sigma^{1/2}$ sandwich structure. For commutative algebras, R must be the Bayesian inverse $R(N, p)(q)_i = p_i N_{j|i}/(\sum_k p_k N_{j|k})$, which in matrix form is $p^{1/2} N^T (Np)^{-1/2}(\cdot)(Np)^{-1/2} N p^{1/2}$. By the functoriality axiom (3) and the Stinespring dilation theorem, this extends uniquely to the noncommutative setting as Eq. (2). The full proof is in Ref. [6]; see also [9] for the Bayesian framework and Supplemental Material [34] for a self-contained derivation. □

Quantum eraser as Petz recovery.—In the quantum eraser, a signal-idler pair is prepared in the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where $|A\rangle, |B\rangle$ are path states and $|d_A\rangle, |d_B\rangle$ are orthogonal detector states. The which-path channel is $\mathcal{N} = \mathrm{Tr}_E$, giving the fully decohered signal state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. Applying the Petz recovery (2) with $\sigma = |\psi\rangle\langle\psi|_{SE}$ and $\mathcal{N} = \mathrm{Tr}_E$:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}, \tag{4}$$

which is exact because Tr_E saturates the DPI for pure bipartite states [34]. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of this Petz recovery. Strictly, the idler measurement is a *quantum instrument* (a collection of CP maps summing to a CPTP map); conditioned on outcome $\pm$, the effective map on the signal is CP but not trace-preserving. The Petz map $\mathcal{R}_{\sigma,\mathrm{Tr}_E}$ itself is CPTP and recovers the full entangled state (4); the subsequent projection onto $|\pm\rangle$ selects the subensemble where the coherent superposition $|A\rangle \pm |B\rangle$ is restored. The functoriality axiom (3) corresponds to the sequential structure of the eraser: composing the which-path marking with the idler measurement decomposes as retrodiction from the idler, then from the path.

Connection to TSVF.—The Aharonov–Bergmann–Lebowitz two-state vector formalism [14] assigns both a forward-evolving state $|\psi\rangle$ and a backward-evolving state $\langle\phi|$. Theorem 1 suggests a canonical choice for the backward state: the Petz-retrodicted state.

FROM RETRODICTION TO ERROR CORRECTION

The uniqueness of the Petz map as retrodiction functor connects naturally to its role as a QEC decoder. The following collects known results; our contribution is the interpretive link.

Theorem 2 (Near-Optimal QEC via Petz Recovery (Barnum–Knill [5], Junge et al. [11])). *Let $\mathcal{N}$ be a noise channel acting on a $[\![n, k, d]\!]$ stabilizer code $\mathcal{C}$ with code projector P and maximally mixed code state $\sigma_{\mathcal{C}} = P/\dim\mathcal{C}$. Let $\epsilon_{\mathrm{KL}} = \max_{a,b} \left\| P E_a^\dagger E_b P - c_{ab} P \right\|$ measure deviation from the Knill–Laflamme conditions [10], where $\{E_a\}$ are Kraus operators of $\mathcal{N}$. Then:*

(a) *If $\epsilon_{\mathrm{KL}} = 0$ (exact QEC), $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on $\mathcal{C}$.*

(b) *In general, the universal recovery theorem of Junge et al. [11] (Corollary 5.1 therein) guarantees the existence of a* rotated *Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ such that:*

$$F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho))\big)^2 \geq \exp\big(-\Delta D\big), \tag{5}$$

where $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ is the relative entropy decrease. The standard Petz map satisfies the weaker Barnum–Knill bound [5]: $F_e \geq \frac{1}{2} F_{e,\mathrm{opt}}^2$.

Proof.—(a) When the Knill–Laflamme conditions hold exactly, the DPI is saturated for all code states: $D(\rho\|\sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ for all ρ supported on $\mathcal{C}$. By Petz's sufficiency theorem [4], the Petz recovery is exact. (b) The rotated Petz map is $\tilde{\mathcal{R}}(\cdot) = \int dt\; p(t)\, \sigma^{it} \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)^{-it}(\cdot)\mathcal{N}(\sigma)^{it}) \sigma^{-it}$, where $p(t) = (\pi/2)(\cosh(\pi t) + 1)^{-1}$; the bound follows from Ref. [11]. □

In this light, QEC decoding is retrodiction: the decoder retrodicts the original ρ from the corrupted $\mathcal{N}(\rho)$ using the code state as prior. The Petz map is the unique Bayesian-consistent choice (Theorem 1), and its near-optimality [5] follows from this uniqueness rather than from case-specific linear algebra.

THE THERMODYNAMIC STRUCTURE

When the reference state is thermal, known results combine into a chain linking recovery fidelity to entropy

production. We state this chain to establish the connection to the temporal asymmetry parameter τ.

Theorem 3 (Master Inequality Chain (assembling [11–13, 18])). *For any quantum channel $\mathcal{N}$, reference state σ, and state ρ:*

$$\boxed{-\log F^2 \;\le\; I(A;E|B)_\omega \;\le\; \Sigma \;\le\; \Delta D} \tag{6}$$

where $F = F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ is the recovery fidelity; $I(A;E|B)_\omega$ is the conditional mutual information of the Stinespring complementary output; Σ is the entropy production (Kolchinsky–Wolpert decomposition [13]); and $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the relative entropy decrease. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state:

$$\Delta D = \beta\, W_{\rm diss} = \Delta S_{\rm irr}/k_{\rm B}, \tag{7}$$

yielding a thermodynamic bound on recovery: $F^2 \ge e^{-\Delta S_{\rm irr}/k_{\rm B}}$.

Proof.—Each step uses a known result; our contribution is the assembly into a single chain with the τ interpretation. Here F denotes the fidelity with the rotated Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ of Ref. [11].

Step 1 (Fidelity bound): By the universal recovery theorem [11], $F^2 \ge \exp(-\Delta D)$, so $-\log F^2 \le \Delta D$.

Step 2 (CMI bound): By Fawzi–Renner [12], $I(A;E|B) \ge -\log F^2$ via the strengthened data processing inequality applied to the Stinespring dilation, giving the leftmost bound.

Step 3 (Entropy production): For a thermodynamic process with Gibbs reference state $\sigma = e^{-\beta H}/Z$, the entropy production satisfies $\Sigma \ge I(A;E|B)$ because the Landauer component $\Sigma_{\rm Landauer} = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ equals $\Delta D \ge I(A;E|B)$ (by Steps 1–2), and $\Sigma \ge \Sigma_{\rm Landauer}$ by the Kolchinsky–Wolpert decomposition [13].

Step 4 (Second law): The relative entropy decrease $\Delta D \ge \Sigma$ by Reeb–Wolf [18]: the free-energy dissipation exceeds the entropy production by system-bath correlations and bath displacement.

The tightness conditions are physically transparent: $F^2 = 1$ iff exact QEC (Knill–Laflamme satisfied); $I(A;E|B) = 0$ iff Markov condition (no information leaks to environment); $\Sigma = 0$ iff thermodynamically reversible process. See Supplemental Material [34] for the detailed derivation including the Gibbs specialization (7). □

Quantum Crooks connection.—When $\sigma = \sigma_\beta = e^{-\beta H}/Z$, the Petz map becomes $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ with the structural factor $\sigma_\beta^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ —exactly the "half Boltzmann weight" appearing in the KMS condition of modular theory. Kwon and Kim [7] proved that this $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ is precisely the reverse channel in the quantum Crooks fluctuation theorem:

$$\frac{p_F(w)}{p_R(-w)} = e^{\beta(w-\Delta F)}, \tag{8}$$

where $p_F(w)$ and $p_R(-w)$ are forward and Petz-reversed work distributions. The detailed balance condition $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ characterizes thermodynamically reversible processes—those that are their own Petz reversal.

Temporal asymmetry parameter.—Define

$$\tau \;=\; 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big). \tag{9}$$

This parameter quantifies the degree to which the channel $\mathcal{N}$ distinguishes past from future when σ is the reference state. A remark on the two Petz maps is in order. The retrodiction uniqueness theorem [6] singles out the *standard* Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ as the unique retrodiction functor; the quantum Crooks theorem likewise uses $\mathcal{R}_{\sigma,\mathcal{N}}$ as the physical reverse channel. The bound (5) and the definition of τ require the *rotated* map $\tilde{\mathcal{R}}$, which averages $\mathcal{R}$ over modular automorphisms $\sigma^{it}(\cdot)\sigma^{-it}$. For a thermal reference $\sigma = e^{-\beta H}/Z$, the rotation $\sigma^{it} = e^{-it\beta H}/Z^{it}$ is simply unitary time evolution generated by H, so $\tilde{\mathcal{R}}$ smooths the standard Petz map over the system's own Hamiltonian dynamics. This thermal averaging handles quantum noncommutativity $[\rho,\sigma] \ne 0$ and ensures that the fidelity bound $F^2 \ge \exp(-\Delta D)$ holds universally. When $[\rho,\sigma] = 0$ (classical limit) or $\Delta D = 0$ (exact recovery), $\tilde{\mathcal{R}}$ reduces to $\mathcal{R}$ and the distinction vanishes. Three properties follow from the master chain (6):

(i) *Closed systems:* If $\mathcal{N}$ is unitary, $\mathcal{R}_{\sigma,\mathcal{U}} = \mathcal{U}^\dagger$ and $F = 1$, so $\tau = 0$. There is no temporal asymmetry.

(ii) *Bound by entropy production:* From $F^2 \ge \exp(-\Sigma)$,

$$\tau \;\le\; 1 - \exp(-\Sigma/2). \tag{10}$$

When $\Sigma = 0$ (zero entropy production), $\tau = 0$ is enforced.

(iii) *Equivalence chain:* For faithful σ (see [34], Sec. S11):

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \text{QMC}, \tag{11}$$

where QMC denotes the quantum Markov chain condition [12]. Each individual equivalence is established [4, 12, 29]; the full chain (11) states them as a single result, formalizing the physical content: the arrow of time ($\tau > 0$) is equivalent to environmental coupling ($I(A;E|B) > 0$), entropy production ($\Sigma > 0$), and imperfect recoverability ($F < 1$).

NEW RESULT: CHANNEL-DEPENDENT BOUND SATURATION

We present a new result that demonstrates a physically meaningful distinction between channel families within the τ framework.

Proposition (Amplitude damping saturation).—Consider single-qubit channels with reference state $\sigma = I/2$. For the amplitude damping channel $\mathcal{N}_\gamma^{\rm AD}$ with damping parameter $\gamma \in [0,1]$:

(a) For the fixed-point input $\rho = |0\rangle\langle 0|$ (ground state, preserved by the channel):

$$F^2 = \frac{1}{1+\gamma} = \exp(-\Delta D), \tag{12}$$

where $\Delta D = \ln(1+\gamma)$. The Junge et al. bound is *exactly saturated* for all γ.

(b) For unital channels (dephasing, depolarizing) with Bloch contraction $\lambda \in (0,1)$, evaluated on equatorial pure states:

$$\eta \equiv \frac{F^2}{\exp(-\Delta D)} = \frac{1+\lambda^2}{2\exp(-H_{\text{bin}})} > 1, \tag{13}$$

where $H_{\text{bin}} = H((1+\lambda)/2)$ is the binary entropy. The bound is *strictly non-saturated* in the interior.

Proof.—For $\sigma = I/2$, the Petz map reduces to $\mathcal{R}(\cdot) = \mathcal{N}^\dagger(\mathcal{N}(I/2)^{-1/2}(\cdot)\mathcal{N}(I/2)^{-1/2})/2$. For unital channels, $\mathcal{N}(I/2) = I/2$ and $\mathcal{R} = \mathcal{N}^\dagger = \mathcal{N}$ (self-adjoint). The fidelity and relative entropy drop are computed explicitly in [34], Sec. S16.

For amplitude damping, $\mathcal{N}(I/2) = (I + \gamma Z)/2$. Direct calculation gives $\mathcal{R}(|0\rangle\langle 0|) = \text{diag}(1/(1+\gamma), \gamma/(1+\gamma))$, yielding $F^2 = 1/(1+\gamma)$, while $\Delta D = \log 2 - D(|0\rangle\langle 0| \| \mathcal{N}(I/2)) = \ln(1+\gamma)$. Hence $F^2 = \exp(-\Delta D)$. □

The physical content: for amplitude damping of the ground state, the temporal asymmetry $\tau = 1 - 1/\sqrt{1+\gamma}$ *exactly equals* the information-theoretic lower bound $1 - \exp(-\Delta D/2)$. There is no slack—the arrow of time is entirely determined by the information loss ΔD. This contrasts with unital channels, where the actual τ is always less than the bound permits.

This distinction is experimentally relevant: amplitude damping is the dominant noise channel in superconducting qubits and ion traps, and Petz recovery has been implemented on both platforms [32, 33]. The prediction $F^2 = 1/(1+\gamma)$ for ground-state input is directly testable.

OBSERVATIONS FROM THE τ FRAMEWORK

Two further observations follow from interpreting existing results through τ.

Observation 1: Post-selection as thermodynamic filtering.—For a stabilizer code under noise $\mathcal{N}$, each syndrome s has a data-processing gap $\Delta D(s)$. Since $\Delta D(s)$ depends only on the noise channel and code state, the per-distance scaling of rejection rates should be decoder-independent. Numerical simulations at $d \in \{3, 5, 7, 9, 11\}$ with three decoders yield a coefficient of variation CV = 0.091 across decoders, consistent with this expectation [34]. This is also consistent with the Ising–QEC mapping [21], statistical-mechanical thresholds [20], and scalable post-selection results [19, 22].

Observation 2: Decoder hierarchy as retrodiction approximation.—Since the Petz map is the unique Bayesian-consistent retrodiction (Theorem 1), all good decoders must approximate it. The known decoder hierarchy [23]—ML > neural network [24] > MWPM > Union-Find—is consistent with being ordered by retrodiction fidelity. A direct test would require computing the channel divergence $D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}} \| \mathcal{D})$ for each decoder $\mathcal{D}$, which we leave to future work.

THE QUANTUM ERASER REVISITED

The quantum eraser is a pure bipartite system: $\mathcal{N} = \text{Tr}_E$, $\sigma = |\psi\rangle\langle\psi|_{SE}$. As computed in [34] (Sec. S1), the Petz map achieves $F = 1$, hence $\tau = 0$. This is a direct consequence of the equivalence chain (11): for a pure bipartite state, $I(A; E|B) = 0$ (no environment beyond the idler) and $\Sigma = 0$ (no dissipation). The quantum eraser "works" because the signal-idler pair forms a closed system with $\tau = 0$.

Superposition necessity.—By contrapositive: if a system is in a definite path state, the Petz recovery produces no coherence (see [34], Sec. S5). Recovery of interference implies prior superposition.

Wheeler's delayed choice and τ.—In Wheeler's delayed-choice experiment [15], the signal-idler pair forms a time-symmetric ($\tau = 0$) system where forward and reverse processes are statistically indistinguishable. The *order* of measurements is operationally irrelevant: measuring the signal first or the idler first yields identical statistics. In the τ language, the delayed-choice aspect requires no retrocausal explanation: the closed system has $\tau = 0$, so the temporal order of measurements is irrelevant—a conclusion consistent with the standard decoherence perspective [27]. The "need to wait" for the idler measurement arises only in the open-system description, where one traces out the idler and observes $\tau > 0$ (decoherence) [27].

DISCUSSION

The interpretive framework presented here rests on known mathematical results—the Petz map's uniqueness as retrodiction functor [6], the universal recovery bound [11], and the connection to entropy production [7, 13, 26]. Our contribution is twofold: the definition of $\tau = 1 - F$ as a continuous measure of the arrow of time, and the demonstration that the amplitude damping channel saturates the recovery bound for fixed-point states.

The τ parameter provides a unified language for phenomena across quantum information:

Domain	**Key result**	τ
Quantum eraser	$F = 1$ (exact)	0
Closed unitary	Petz = inverse	0
QEC (Ref. [11])	$F^2 \geq e^{-\Delta D}$	> 0
Thermodynamics	Eq. (10)	$\leq 1 - e^{-\Sigma/2}$
Ampl. damping	Eq. (12) (new)	$= 1 - e^{-\Delta D/2}$

The physical picture connects to the decoherence program [27, 28]: environment-induced decoherence produces entropy ($\Sigma > 0$), implying $\tau > 0$. The arrow of time, in this interpretation, is a consequence of system-environment coupling quantified by Petz recovery failure.

Limitations.—The τ bound $\tau \leq 1 - \exp(-\Sigma/2)$ has not yet been tested by simultaneous measurement of F and Σ on the same quantum channel. The amplitude damping saturation (Eq. (12)) is a single-qubit result; extension to multi-qubit channels remains open. The QEC observations (Sec.) are numerically supported but lack rigorous proofs.

Open questions.—(i) For non-Markovian processes, Σ can be temporarily negative [30], implying transient decreases in τ. Can the process tensor framework [16] extend the equivalence chain (11)? (ii) Does the saturation result (12) generalize to multi-qubit amplitude damping or other non-unital channels? (iii) Since entanglement wedge reconstruction uses the Petz map [8], does τ have a holographic interpretation?

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[34] See Supplemental Material for detailed derivations, the explicit quantum eraser calculation, the amplitude damping saturation proof, post-selection analysis, the equivalence chain proof, and caveats.